

Phil meeting notes

December 30, 2025

Meeting Notes

- Discussed current status of the project
 - Multiple Tool Usage
 - April/Aruco Tag based motion
 - Bluesky based robot controls
 - Hardcoded positions
- Github write access provided to Kushant

Tagless Detection

- Currently orientation is assumed to be constant (flat and perpendicular to camera)
- Orientation accuracy needs to be tested if not flat
- If using a Hand-E gripper, requires accurate COM and orientation to properly grasp objects

VM Camera Communication

- VM cannot communicate with camera
- Camera can be configured in the same subnet as robot instead of CAM subnet (consider camera as an extension of the robot)

Bluesky - ROS Integration

ROS - Bluesky : For experiment number / equipment position changes / don't move during measurement:

- Redis database service or CSV export or NEXT 3 has sample database (coming up)
- Dynamic Objects : Read EPICS PVs in ROS to get current positions of equipment and update planning scene accordingly (PyEPICS to query PVs?)

Integration with VISION / Bluesky Adaptive

- Bluesky Adaptive - framework to run (custom) algorithms to decide which sample to measure next based on previous measurements.
- Bluesky Adaptive → Bluesky → Redis Lookup → sends to ROS (run autonomous Bluesky plans)

Containerization Strategy

- Containerized separation of Bluesky and ROS is to ensure ROS doesn't have the rights to access / interact with beamline data / controls
- EPICS and ROS are UDP based — might cause network overhead?

CDS Collaboration

- CDS — research. NSLS — implementation that works
- Vision language action models?
- Use resources built by CDS — for example models for grasping objects and fine tune them to work at beamlines

Things to Present

1. What is currently possible + what is being blocked
2. Exact specifications required
3. Showcase experiments that couldn't be done before can be done now